Thesis…

THESIS STRUCTURE FACULTY OF COMPUTER SCIENCE & IT

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Thesis topic selection

The supervisor should ask the students the following questions, to decide if the topic is acceptable or not:

- Does the topic align with the thesis guidelines?
- The topic should not be taken earlier.
- Are you interested in this topic?
- Can you find enough information on this topic?
- Can you find the types of resources you need?
- Can you effectively research this topic within the deadlines?

Contribution

In a thesis, the research contribution is the original and novel aspect of the research that adds new knowledge to the field. It can be a new theory, a new methodology, a new empirical finding, or a new application of existing knowledge.

The significance of a thesis should portray the general and specific contribution and application of the study, the grounds on which the study is based, and the benefits of the study.

The contribution of a thesis depends on the questions answered and the theories and methods used.

Preface pages

- Title Page
- Original Literary Work Declaration
- Abstract
- Acknowledgements
- Table of Contents
- List of Figures
- List of Tables
- List of Symbols and Abbreviations
- List of Appendices

Thesis Chapters

The supervisor should ensure that the Cs/IT students' dissertations contain the following chapters:

1. **Chapter One**: Introduction
    - 1.1 Background
    - 1.2 Problem Statement
    - 1.3 Research Objectives
    - 1.4 Research Questions
    - 1.5 Motivation of the study
    - 1.6 Significance of the Study
    - 1.7 Scope of study
    - 1.8 Organization of the study
    - 1.9 Key Terms
2. **Chapter Two:** Literature Review
    - 2.1 Introduction
    - 2.2 Theoretical Framework / Models
    - 2.3 Related Work (Past Studies / Existing Systems)
    - 2.4 Critical Review & Gap Identification
3. **Chapter Three:** Methodology
    - 3.1 Introduction
    - 3.2 Research Design
    - 3.3 System description

        - 3.4 Overview of the system
        - 3.5 System features
        - 3.6 System Development Environment
        - 3.7 System Requirement Specification
            - 3.7.1 Hardware Requirements
            - 3.7.2 Software Requirements

N.B: if the system related dataset like machine learning and etc, the chapter three should contain the following points:

    - 3.1 Introduction
    - 3.2 Research Design
    - 3.3 System description
    - 3.4 System Architecture
    - 3.5 System features
    - 3.6 Methodology
    - 3.7 Data Collection
    - 3.8 Data Preparation
    - 3.9 System Requirement

4. **Chapter Four:** Analysis and design
    - 4.1 System analysis
    - 4.2 Existing Approaches
    - 4.3 The Proposed System
    - 4.4 Requirements
        - 4.4.1 Functional Requirements
        - 4.4.2 Nonfunctional requirements
    - 4.5 Feasibility Study
        - 4.5.1 Technical Feasibility
        - 4.5.2 Economic Feasibility
        - 4.5.3 Schedule feasibility
    - 4.6 System Design
        - Data Flow Diagrams ((DFD, UML, Use Case Diagrams, ER Diagram)
    - 4.7 Database Design
    - 4.8 System Architecture Diagram
5. **Chapter Five:** Implementation & Testing
    - 5.1 Implementation Environment
    - 5.2 Coding & Modules

    - 5.3 Testing Strategy
    - 5.4 Test Cases & Results
    - 5.5 Validation & Verification
6. **Chapter Six:** Results and Discussion
    - 6.1 Results Presentation
    - 6.2 Comparison with Existing Systems
    - 6.3 Evaluation Metrics
    - 6.4 Discussion of Findings
7. **Chapter Seven:** Conclusion & Future Work.
    - 7.1 Summary of Key Findings
    - 7.2 Conclusion
    - 7.3 Recommendations
    - 7.4 Future Work
8. **References / Bibliography**
    - (APA/IEEE/Oxford depending on faculty rule).
9. **Appendices**
    a. Source Code
    b. Additional Diagrams
    c. Questionnaires / Data Collection Tools
    d. Raw Data

Research papers

- Each group is required to submit a research paper when they finish the book.
- Use Font format previous requirements

Thesis Structure with Descriptions

Chapter One: Introduction

This chapter introduces the research problem, context, and purpose of the study. It sets the foundation of the thesis.

- **1.1 Background**
  Explains the general area of study, trends, and developments. Provides context and rationale for why the topic is important.
- **1.2 Problem Statement**
  Clearly defines the problem or challenge the research aims to solve. It should highlight the gap in existing knowledge or systems.
- **1.3 Research Objectives**
  States what the study intends to achieve. Objectives should be SMART (Specific, Measurable, Achievable, Realistic, Time-bound).
- **1.4 Research Questions**
  Lists the main and sub-research questions the study seeks to answer.
- **1.5 Motivation of the Study**
  Justifies why the researcher is interested in the topic—academic, professional, or societal relevance.
- **1.6 Significance of the Study**
  Explains the contributions of the study (theoretical, practical, and technological).
- **1.7 Scope of the Study**
  Defines the boundaries of the research (what is covered and what is not). Mentions constraints like time, resources, or dataset size.
- **1.8 Organization of the Study**
  Provides an outline of the thesis structure (what each chapter covers).

Chapter Two: Literature Review

This chapter critically reviews existing research, theories, and systems related to the study.

- **2.1 Introduction**
  Briefly explains the purpose of the literature review and its structure.

- **2.2 Theoretical Framework / Models**
  Discusses theories, models, or conceptual frameworks relevant to the topic.

- **2.3 Related Work (Past Studies / Existing Systems)**
  Reviews prior studies, systems, or methodologies. Shows what has been done and how it relates to the current research.

- **2.4 Critical Review & Gap Identification**
  Summarizes limitations or gaps in previous work and positions the current research as a way to fill those gaps.

Chapter Three: Methodology

This chapter explains how the research is conducted, including tools, techniques, and processes.

- **3.1 Introduction**
  Introduces the methodology chapter and research approach.

- **3.2 Research Design**
  Describes whether the research is experimental, qualitative, quantitative, mixed methods, or system development.

- **3.3 System Description**
  Provides an overview of the system or solution being developed.

- **3.4 Overview of the System**
  Explains main modules, features, and functionality.

- **3.5 System Features**
  Lists system capabilities (e.g., user management, data processing, analytics).

- **3.6 System Development Environment**
  Describes tools, platforms, and programming languages used (e.g., Python, Java, SQL, TensorFlow).

- **3.7 System Requirement Specification (SRS)**
  - **3.7.1 Hardware Requirements** – hardware specifications (servers, PCs, GPU, etc.).
  - **3.7.2 Software Requirements** – operating systems, software tools, IDEs, libraries.

N.B. For dataset/system-related research (like AI/ML projects):

- **3.3 System Description** – overview of dataset/system.
- **3.4 System Architecture** – diagram of data flow and components.
- **3.5 System Features** – expected functionalities.
- **3.6 Methodology** – algorithm selection, model training, techniques.
- **3.7 Data Collection** – sources of data.
- **3.8 Data Preparation** – preprocessing, cleaning, feature engineering.
- **3.9 System Requirement** – hardware/software needs.

Chapter Four: Analysis and Design

Explains how the problem was analyzed and how the system is designed.

- **4.1 System Analysis**
  Identifies needs and requirements for the system.
- **4.2 Existing Approaches**
  Reviews existing solutions, their strengths and weaknesses.
- **4.3 The Proposed System**
  Introduces the new system and how it improves on existing ones.
- **4.4 Requirements**
  - **4.4.1 Functional Requirements** – core functions (e.g., login, report generation).
  - **4.4.2 Non-functional Requirements** – performance, security, usability.
- **4.5 Feasibility Study**
  Evaluates technical, economic, and operational feasibility of the system.
- **4.6 System Design**
  Includes diagrams such as:
  - Data Flow Diagrams (DFD)
  - UML diagrams
  - Use Case Diagrams

    - ER Diagrams

- **4.7 Database Design**
  Describes schema, tables, relationships, and constraints.

- **4.8 System Architecture Diagram**
  Provides a high-level view of how system components interact.

Chapter Five: Implementation & Testing

Details the actual building of the system and validation of its performance.

- **5.1 Implementation Environment**
  Hardware, OS, and software setup.

- **5.2 Coding & Modules**
  Description of major modules and their functionalities.

- **5.3 Testing Strategy**
  Testing methods (unit, integration, system, user acceptance testing).

- **5.4 Test Cases & Results**
  Tables showing input, expected output, actual output.

- **5.5 Validation & Verification**
  Confirms the system meets requirements and performs correctly.

Chapter Six: Results and Discussion

Presents and interprets the findings.

- **6.1 Results Presentation**
  Tables, graphs, charts showing outcomes.

- **6.2 Comparison with Existing Systems**
  Benchmarks performance against other systems.

- **6.3 Evaluation Metrics**
  Measures such as accuracy, precision, recall, response time, usability ratings.

- **6.4 Discussion of Findings**
  Explains meaning of results and answers the research questions.

Chapter Seven: Conclusion & Future Work

Wraps up the thesis and suggests further improvements.

- **7.1 Summary of Key Findings**
  Recap of main results.

- **7.2 Conclusion**
  Overall statement of achievement.

- **7.3 Recommendations**
  Suggestions for practical use of findings.

- **7.4 Future Work**
  Areas for further research and system improvement.

References / Bibliography

- List of all sources cited (APA, IEEE, or Oxford style as required by faculty).

Appendices

- **a. Source Code** – full or partial code listing.
- **b. Additional Diagrams** – supporting visuals not included in main body.
- **c. Questionnaires / Data Collection Tools** – survey forms, interview guides.
- **d. Raw Data** – datasets or logs.

END